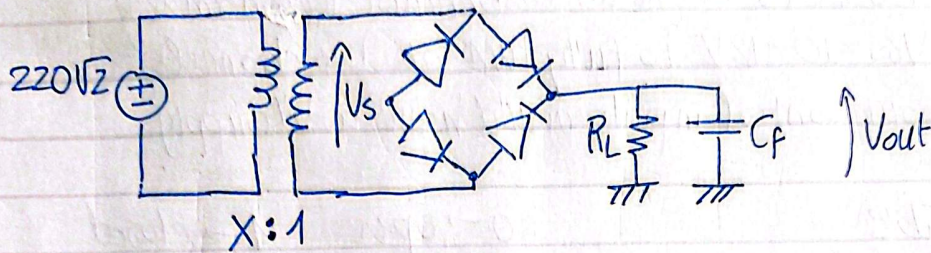


Task 2.1

a) 220RMS $\rightarrow$ 12VDC



$$V_s = 12V + \underbrace{1.4V}_{\text{Diode Bridge Drop}} + \underbrace{1.5V}_{\text{Ripple Allowance}} = 14.9V$$

$$\text{Ratio } \frac{220\sqrt{2}}{14.9} = \frac{x}{1} \quad x = \underline{\underline{21}}$$

$$V_{\text{ripple}} = 1.5 = \frac{I}{f \times C_f} = \frac{1A}{100\text{Hz} \times C_f} \quad C_f = 6.66\text{mF}$$

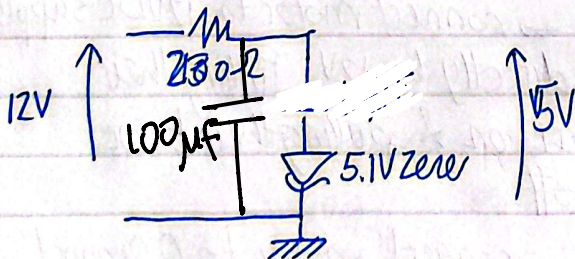
$$f = 50 \quad \text{1 cycle}$$

$$f = 100 \quad \text{2 cycles}$$

$$r\% = \frac{2}{4fR_L C_f - 1}$$

$$r\% = \frac{1.5}{12} = 0.125 = \frac{2}{4(50)R_L(6.66\text{m}) - 1} \quad R_L = 12.87\Omega \approx 13\Omega$$

b) 220RMS $\rightarrow$ 5V $\Rightarrow$ RC filter + Zener (low noise)



$$R = \frac{V_{in} - V_z}{I_R} = \frac{12 - 5.1}{30\text{mA}} = 230\Omega$$

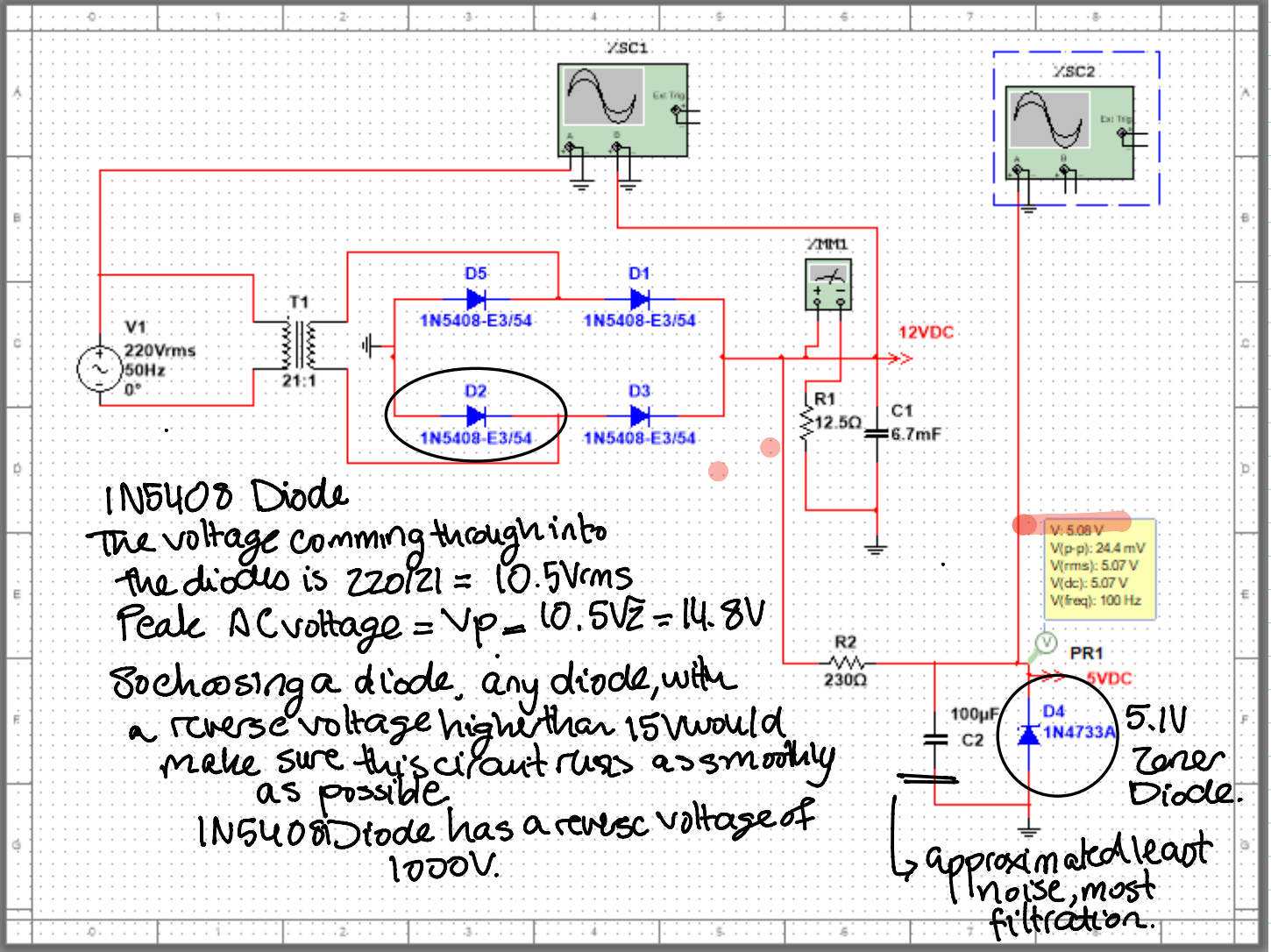
$\hookrightarrow$ Assumption

$$I_L = 20\text{mA}$$

$$I_z = 10\text{mA}$$

Capacitor $\rightarrow$ test for noise cancellation
with bigger values
 $C = 100\mu\text{F}$

Task 2#1



Oscilloscope 1:
 Channel A: 220Vrms AC voltage (input)
 Channel B: 12VDC voltage (rectified)



Oscilloscope 2:
 Channel A: 5VDC low noise.

Task 2.2

Power Supply: 12VDC

Control inputs: 5VDC via 2 switches (S_1 & S_2)

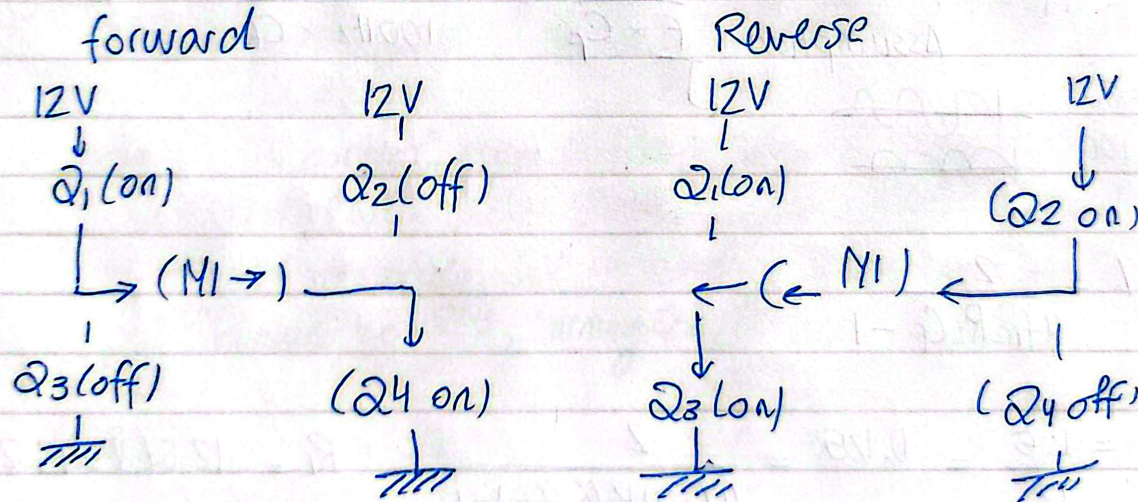
We avoid MOSFETs (logic level) because they normally require $V_{GS} = 10-12V$ to fully turn on the channels.

But our control inputs would have a 5V supply.

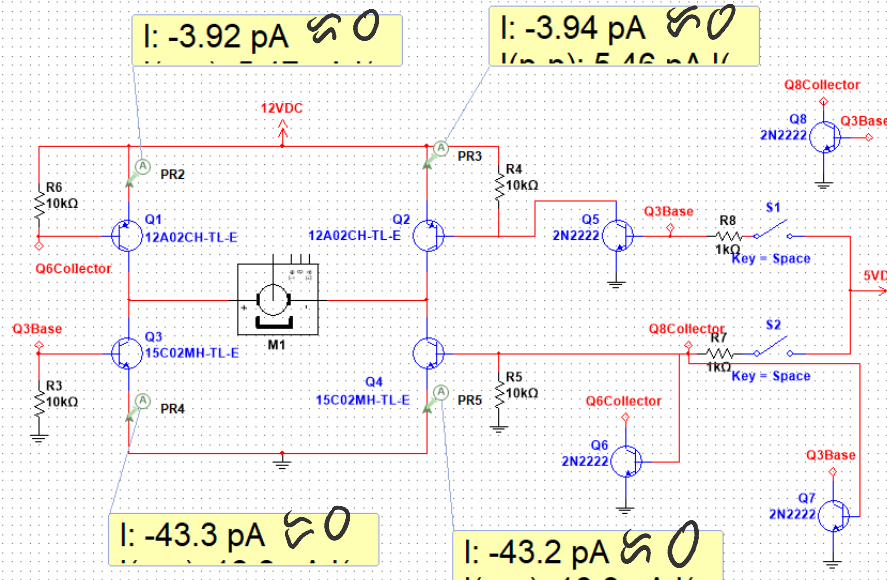
Logic Table

0 $\rightarrow$ open 1 $\rightarrow$ closed

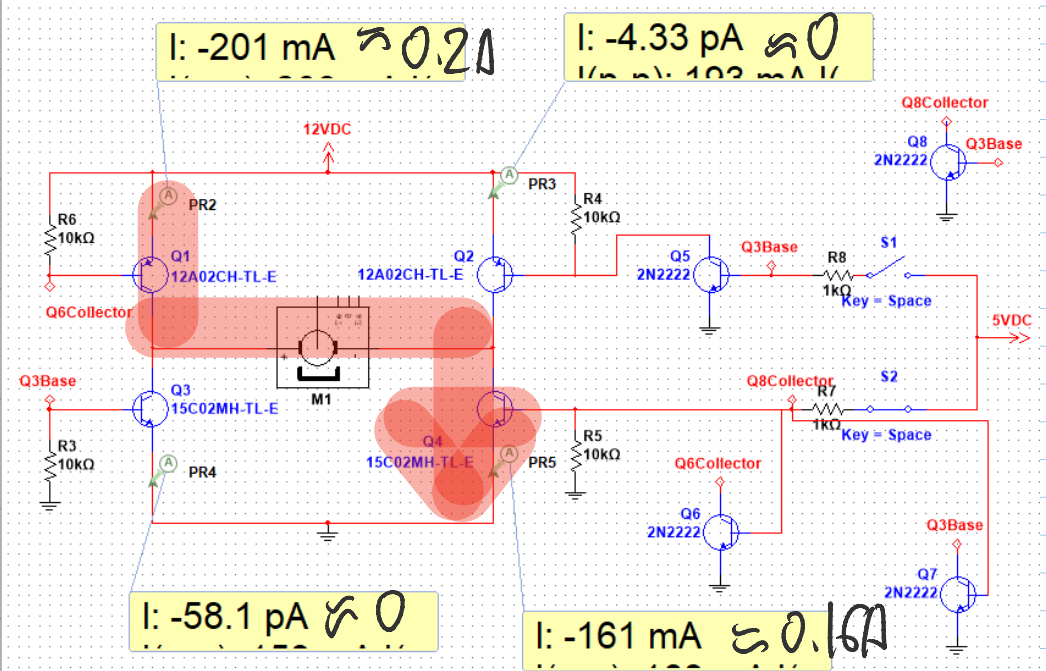
S_1	S_2	
0	0	No movement
0	1	forward (Direction 1)
1	0	reverse (Direction 2)
1	1	No movement



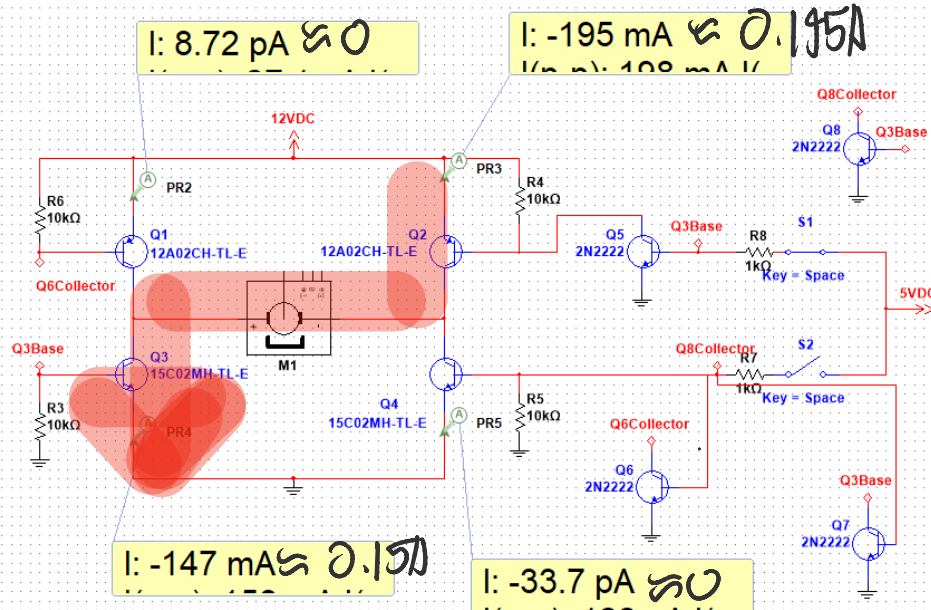
S_1/S_2 open (no movement)



S_1 open, S_2 closed (forward)



S_1 closed, S_2 open (reverse)



S_1 & S_2 closed (no movement)
but its not working very properly,
tried to
debug it

by adding
switch like
 Q_7 & Q_8
which improved
but not
entirely fixed (it works like forward)

